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A Bayesian Step Least Mean Squares Algorithm for Gaussian Signals

Paulo Alexandre Crisóstomo Lopes

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Group: 26

Member： 曾遠騏 N26120498

黃御倫 N26121795

李宗祐 N26121842

Outline

1 Introduction

2 Algorithm Derivation

3 Simulation Results

4 Conclusion

Introduction

Some results show that the LMS is equivalent to the Kalman filter (KF) if the state noise of the KF and the step of the LMS algorithm are carefully selected.

Introduction

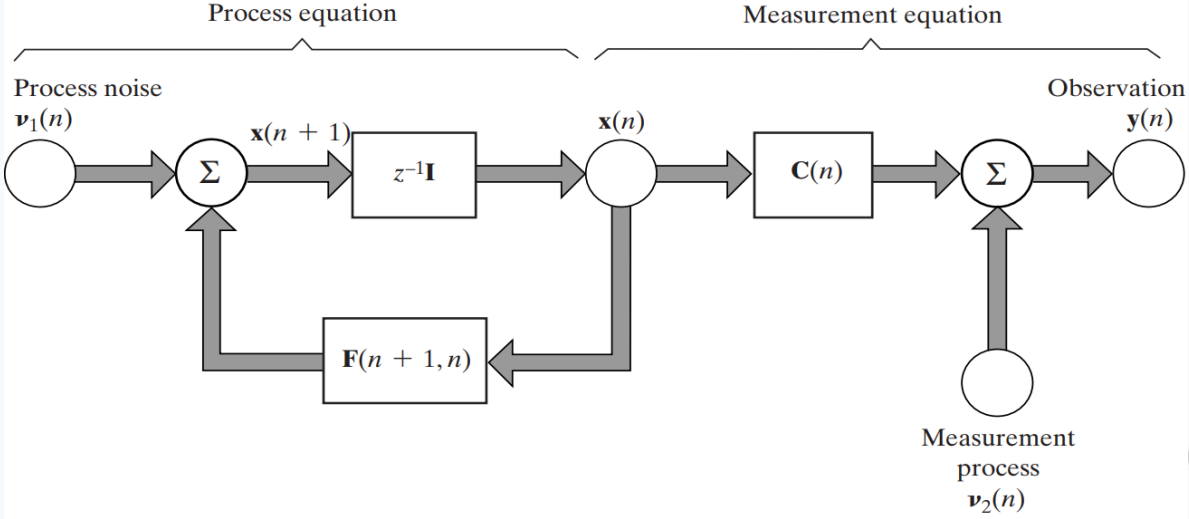
- Selecting the step of the least mean squares (LMS) algorithm is an old problem.
- The LMS algorithm, with time-varying step, size can be shown to be equivalent to the Kalman filter(KF) in some conditions.
- The optimum step size can be calculated using estimates of the probability density function (PDF) of the coefficient estimation error variance (q_w) and measurement noise variance (q_v).

Introduction

- Algorithms such as NLMS,EM,JO-LMS can choose the step size are based on reference is white or known measurement noise.
- [27] use autocorrelation to obtain noise covariance matrices , however, it's assuming that the measurement noise is white.
- [28,29] they derive formulas for the covariance matrix of the measures **C** and invert them to obtain the state noise covariance matrix **Q** assuming the measurement noise matrix **R** is known.

[27]Mehra, R.: 'On the identification of variances and adaptive Kalman filtering',IEEE Trans. Autom. Control, 1970, 15, (2), pp. 175–184
[28] Feng, B., Fu, M., Ma, H., et al.: 'Kalman filter with recursive covariance estimation-sequentially estimating process noise covariance', IEEE Trans.Ind. Electron., 2014, 61, (11), pp. 6253–6263
[29] Wang, H., Deng, Z., Feng, B., et al.: 'An adaptive Kalman filter estimatingprocess noise covariance', Neurocomputing, 2017, 223, pp. 12–17

Introduction



$$\boldsymbol{x}(n+1) = \mathbf{F}(n+1,n)\mathbf{x}(n) + \boldsymbol{v}_1(n) \qquad (process\ equation)$$

$$\boldsymbol{y}(n) = \mathbf{C}(n)\mathbf{x}(n) + \boldsymbol{v}_2(n) \qquad (measurement\ equation)$$

$$\boldsymbol{R}(n) = \boldsymbol{C}(n)\boldsymbol{K}(n+1,n)\mathbf{C}^H(n) + \boldsymbol{Q}_2(n)$$

Algorithm Derivation

The method used in this paper is to use the joint probability density function to find q_w, q_V and use the Gaussian approximation method to get the update equation.

Step Size Calculation[5]

LMS algorithm: $\boldsymbol{w}(n) = \boldsymbol{w}(n-1) + \mu\, e(n)\boldsymbol{u}(n)$

$e(n) = d(n) - \boldsymbol{u}^{\boldsymbol{T}}(n)\boldsymbol{w}(n-1)$
$\mathbf{u}(n) = [u(n) \dots u(n-N+1)]^{\boldsymbol{T}}$
$\boldsymbol{w}(n) = [w_0(n) \dots w_{N-1}(n)]^{\boldsymbol{T}}$

Step size from: $\mu(n) = \dfrac{1}{p_t(n) + q_v(n)/q_w(n)}$

$p_t(n) = \boldsymbol{u}^{\boldsymbol{T}}(n)\mathbf{u}(n)$
q_v: *measurement noise variance*
q_w: *variance of the current estimate of each coefficient*

Optimum coefficient: $\boldsymbol{w}(n) = \mathrm{E}[\boldsymbol{x}(n) | q_w(n), q_v(n), \boldsymbol{w}(n-1)]$

$\boldsymbol{x}(n) = \boldsymbol{w_0}(n)$: *optimal values*

for a carefully chosen state noise

[5] Lopes, P.A., Gerald, J.B.: ‘New normalized LMS algorithms based on the Kalman filter’. IEEE Int. Symp. on Circuits and Systems, 2007. ISCAS 2007.IEEE, New Orleans, LA, USA, 2007, pp. 117–120

Step Size Calculation

- $q_w(n)$ and $q_v(n)$ are unknown

➢ $$\boldsymbol{w}_b(n) = \mathrm{E}[\boldsymbol{x}(n)|\boldsymbol{w}(n-1)]$$
$$= \int\int^{\mathrm{E}}[\boldsymbol{x}(n)|q_w(n), q_v(n), \boldsymbol{w}(n-1)]$$
$$\rho_q(q_w(n), q_v(n), n)\mathrm{d}q_w(n)\mathrm{d}q_v(n)$$

$\rho_q(q_w(n), q_v(n), n)$: *the joint PDF of* $q_w(n)$ *and* $q_v(n)$

Law of total probability:

$$P(A) = \int P(A|B) \cdot P(B)\,\mathrm{d}B$$

➢ $$\boldsymbol{w}_b(n) = \int\int^{\boldsymbol{w}}(n)\rho_q(q_w(n), q_v(n), n)\mathrm{d}q_w(n)\mathrm{d}q_v(n)$$

➢ $$\boldsymbol{w}_b(n) = \boldsymbol{w}_b(n-1) + \mu_b(n)\, e(n)\boldsymbol{u}(n)$$

$$\mu_b(n) = \mathrm{E}[\mu(n)] = \int\int \mu(n)\rho_q(q_w(n), q_v(n), n)\mathrm{d}q_w(n)\mathrm{d}q_v(n)$$

Since only $\mu(n)$ depends on $q_w(n)$ *and* $q_v(n)$

State Variables PDF

- Calculate $q_w \rho_q(q_w(n), q_v(n), n)$

$\boldsymbol{q}(n) = [q_w(n), q_v(n)]^{\boldsymbol{T}}$
$\boldsymbol{a}(n) = [p_t(n), \mathbf{1}]^{\boldsymbol{T}}$
$E[\mathrm{e}^2(\mathrm{n})] = \boldsymbol{a}^{\boldsymbol{T}}(n)\boldsymbol{q}(n)$
$\boldsymbol{e}(n)$:zero-mean Gaussian

- $$\rho_{\boldsymbol{q}(n)|n}(\boldsymbol{q}) = \rho_{\boldsymbol{q}(n)|e(n),e(n-1),\ldots}(\boldsymbol{q}) = \frac{\rho_{e(n)|\boldsymbol{q}(n)}(\boldsymbol{q})\rho_{\boldsymbol{q}(n)|n-1}(\boldsymbol{q})}{\rho_{e(n)|n-1}(\boldsymbol{q})} = k\frac{\mathrm{e}^{-(e^2(n)/2\boldsymbol{a}^{\mathrm{T}}(n)\boldsymbol{q})}}{\sqrt{2\pi\,\boldsymbol{a}^{\mathrm{T}}(n)\boldsymbol{q}}}\rho_{\boldsymbol{q}(n)|n-1}(\boldsymbol{q})$$

Bayes' rule:

$$f_{X|Y}(x|y) = \frac{f_X(x)\cdot f_{Y|X}(y|x)}{f_Y(y)}$$

- $$\rho_{\boldsymbol{q}|n}(\boldsymbol{q}) = k\left(\prod_{i=1}^{n}\frac{\mathrm{e}^{-(e^2(i)/2\boldsymbol{a}^{\mathrm{T}}(i)\boldsymbol{q})}}{\left((2\pi)^2\boldsymbol{q}^{\mathrm{T}}\boldsymbol{a}(i)\boldsymbol{a}^{\mathrm{T}}(i)\boldsymbol{q}\right)^{1/4}}\right)\rho_{\boldsymbol{q}|0}(\boldsymbol{q})$$

Assume

$$\rho_{\boldsymbol{q}(n)|n-1}(\boldsymbol{q}) = \rho_{\boldsymbol{q}(n)|n-1}(\boldsymbol{q})$$

- $$\rho_{\boldsymbol{q}(n)|n}(\boldsymbol{q}) \simeq \hat{\rho}_{\boldsymbol{q}}(\boldsymbol{q}, \boldsymbol{D}(n), \boldsymbol{c}(n), n) = k_0\frac{\mathrm{e}^{-\left(n/2\boldsymbol{c}^{\mathrm{T}}(n)\boldsymbol{q}\right)}}{\left((2\pi)^2\boldsymbol{q}^{\mathrm{T}}\boldsymbol{D}(n)\boldsymbol{q}\right)^{n/4}}$$

$$\boldsymbol{D}(n) = \begin{bmatrix} d_{1,1}(n) & d_{1,2}(n) \\ d_{2,1}(n) & d_{2,2}(n) \end{bmatrix}$$

$$\mathbf{c}(n) = [c_1(n), c_2(n)]^{\boldsymbol{T}}$$

State Variables PDF

- Calculate $q_w \rho_q(q_w(n), q_v(n), n)$

➢ $$\rho_\mu(z) = \int \frac{\hat{\rho}_{\boldsymbol{q}}(\boldsymbol{q}, \boldsymbol{D}, \boldsymbol{c}, n)}{z^2/q_w} \mathrm{d}q_w$$

$$= \frac{1}{4}\pi^{-n/2}\Gamma\left(\frac{n}{2}-2\right)\left(\frac{-c_2 z p_t + c_1 z + c_2}{n}\right)^{(n-4)/2}$$

$$\left(d_{2,2}(zp_\mathrm{t}-1)^2 + z(zd_{1,1} - 2d_{1,2}(zp_\mathrm{t}-1))\right)^{-n/4}$$

Γ =Gamma function

Change of random variable:

$$\rho_y(y) = \int \rho_x(x) \left|\frac{dy}{dx}\right| \delta(y - g(x))\, dx$$

➢ $$\mathrm{E}[\mu] \simeq \int_0^{1/p_\mathrm{t}} z\hat{\rho}_\mu(z)\mathrm{d}z$$

Approximated by a Gaussian function

➢ $$a = \frac{\sqrt{p_z}(1 - z_1 p_t)}{\sqrt{2}p_t}, \quad b = \frac{z_1\sqrt{p_z}}{\sqrt{2}},$$

$$\mathrm{E}[\mu] \simeq z_1 + \frac{-2\mathrm{e}^{-a^2} + 2\mathrm{e}^{-b^2}}{\sqrt{2\pi p_z}(\mathrm{erf}(a) + \mathrm{erf}(b))}$$

mean: $z_1 = z_0 - \frac{\delta_1(z_0)}{\delta_2(z_0)}$

$variance: q_z = -\frac{1}{\delta_2(z_0)}$

let $p_z = \frac{1}{q_z}$

$erf(x) = Gauss\ error\ function$

Measure Update Equations

- Determine how to update **D** and **c**

$\boldsymbol{A}^{-1}$:multivariate Gaussian covariance matrix
mean:$q_0 = \boldsymbol{A}^{-1}\boldsymbol{b}$

➢ $\hat{\rho}_{\boldsymbol{a}}(\boldsymbol{q}, \boldsymbol{D}, \boldsymbol{c}, n) \simeq \tilde{\rho}_{\boldsymbol{a}}(\boldsymbol{q}, \boldsymbol{A}, \boldsymbol{b}) = k_1 \mathrm{e}^{\boldsymbol{b}^{\mathrm{T}}\boldsymbol{q} - \frac{1}{2}\boldsymbol{q}^{\mathrm{T}}\boldsymbol{A}\boldsymbol{q}}$ ➡ Approximated by a Gaussian function

(i) The values of $\boldsymbol{q}$ that maximizes both functions are equal.

$$\boldsymbol{q}_{\mathrm{o}} = \frac{\boldsymbol{D}^{-1}\boldsymbol{c}}{\boldsymbol{c}^{\mathrm{T}}\boldsymbol{D}^{-1}\boldsymbol{c}} = \boldsymbol{A}^{-1}\boldsymbol{b}$$

(ii) The Hessian matrices of the logarithm of both functions($\ln \widehat{\rho_{\boldsymbol{q}}}$ and $\ln \widetilde{\rho_{\boldsymbol{q}}}$)are equal at the maximum point $\boldsymbol{q_0}$.

$$\frac{n}{2}\boldsymbol{D}\left(\boldsymbol{c}^{\mathrm{T}}\boldsymbol{D}^{-1}\boldsymbol{c}\right) = \boldsymbol{A}$$

Measure Update Equations

- Determine how to update **D** and **c**

➢ $$\boldsymbol{D} = \boldsymbol{A},\ \xi = \boldsymbol{b}^{\mathrm{T}}\boldsymbol{A}^{-1}\boldsymbol{b}$$

$$\boldsymbol{c} = \frac{\boldsymbol{b}}{\xi},\ n = 2\xi\,.$$

$$\xi = \boldsymbol{b}^{\boldsymbol{T}}\boldsymbol{q}_{\boldsymbol{0}} = \boldsymbol{q}_{\boldsymbol{0}}{}^{\boldsymbol{T}}\boldsymbol{A}\boldsymbol{q}_{\boldsymbol{0}}$$
$$\zeta = \boldsymbol{a}^{\boldsymbol{T}}\boldsymbol{q}_{\boldsymbol{0}}$$

➢ $$\rho_{\boldsymbol{q}(n)|n}(\boldsymbol{q}) \simeq k\frac{\mathrm{e}^{-\left(\left(e^2(n)-\boldsymbol{a}^{\mathrm{T}}(n)\boldsymbol{q}(n)\right)^2/\left(4\zeta^2(n)\right)\right)}}{\sqrt{4\pi\,\zeta^2(n)}}\tilde{\rho}_{\boldsymbol{q}}(\boldsymbol{q})$$

mean: $\boldsymbol{a}^{\boldsymbol{T}}(n)\boldsymbol{q}(n)$
variance: $2\zeta^2$

➢ $$\boldsymbol{A}(n+1) = \boldsymbol{A}(n) + \frac{\boldsymbol{a}(n)\boldsymbol{a}^{\mathrm{T}}(n)}{2\zeta^2(n)}$$

$$\boldsymbol{b}(n+1) = \boldsymbol{b}(n) + \frac{\boldsymbol{a}(n)e^2(n)}{2\zeta^2(n)}\,.$$

➡ the measure update equations

Time Update Equations

Table 1 Proposed algorithm

Initialisation

$$\boldsymbol{w}(0) = [1, 0, \ldots, 0]^{\mathrm{T}} \tag{31}$$

$$\boldsymbol{q}_o(0) = [1, 1]^{\mathrm{T}} \tag{32}$$

$$\boldsymbol{A} = 0, \ \boldsymbol{b} = 0 \tag{33}$$

$$\delta = \text{small number} \tag{34}$$

Iterate

$$e(n) = d(n) - \boldsymbol{u}(n)^{\mathrm{T}}\boldsymbol{w}(n) \tag{35}$$

$$p_t(n) = \boldsymbol{u}(n)^{\mathrm{T}}\boldsymbol{u}(n), \ \ p(n) = p_t(n)/N \tag{36}$$

$$\boldsymbol{a}(n) = [p_t(n), 1]^{\mathrm{T}} \tag{37}$$

$$\zeta(n) = \boldsymbol{a}^{\mathrm{T}}(n)\boldsymbol{q}_o(n) \tag{38}$$

$$\boldsymbol{A}(n) = \boldsymbol{A}(n-1) + \frac{\boldsymbol{a}(n)\boldsymbol{a}^{\mathrm{T}}(n)}{2\zeta^2(n)} \tag{39}$$

$$\boldsymbol{b}(n) = \boldsymbol{b}(n-1) + \frac{\boldsymbol{a}(n)e^2(n)}{2\zeta^2(n)} \tag{40}$$

$$\boldsymbol{q}_o(n) = (\boldsymbol{A}(n) + \delta\boldsymbol{I})^{-1}\boldsymbol{b}(n) \tag{41}$$

$$\xi(n) = \boldsymbol{b}(n)^{\mathrm{T}}\boldsymbol{q}_o(n) \tag{42}$$

$$\boldsymbol{c}(n) = \boldsymbol{b}(n)/\xi(n) \tag{43}$$

$$n_{\mathrm{x}}(n) = 2\xi(n) \tag{44}$$

$$[q_w(n), q_v(n)]^{\mathrm{T}} = \boldsymbol{q}_o(n) \tag{45}$$

$$z(n) = \min\left(\max\left(\frac{q_w(n)}{q_w(n)p_t(n) + q_v(n)}, 0\right), \frac{1}{p_t}\right) \tag{46}$$

$$\mu(n) = \begin{cases} z(n) & n_{\mathrm{x}} \le 4 \\ 1/(2p_t) & n_{\mathrm{x}} > 4 \text{ and } \delta_2(n) > 0 \\ \text{calculate (17)} & \text{otherwise} \end{cases} \tag{47}$$

$$\boldsymbol{w}(n+1) = \boldsymbol{w}(n) + \mu(n)\boldsymbol{u}(n)e(n) \tag{48}$$

$$\gamma_1(n) = 1 \quad \text{if} \quad q_w(n) > 0 \quad \text{else} \quad -1 \tag{49}$$

$$\gamma_2(n) = 1 \quad \text{if} \quad q_v(n) > 0 \quad \text{else} \quad -1 \tag{50}$$

$$\boldsymbol{C}(n) = \begin{bmatrix} \gamma_1(n)(1 - \alpha\mu(n)p(n))^2 & \alpha\gamma_2(n)\mu^2(n)p(n) \\ 0 & \gamma_2(n) \end{bmatrix} \tag{51}$$

$$\hat{\boldsymbol{A}}(n+1) = (\boldsymbol{C}(n)^{-1})^{\mathrm{T}}\boldsymbol{A}(n)\boldsymbol{C}(n)^{-1} \tag{52}$$

$$\hat{\boldsymbol{b}}(n+1) = (\boldsymbol{C}(n)^{-1})^{\mathrm{T}}\boldsymbol{b}(n) \tag{53}$$

$$k_{\mathrm{t}}(n) = 1 \text{ if } \mathrm{trace}\left(\hat{\boldsymbol{A}}(n+1)\right) \le T \text{ else } \frac{T}{\mathrm{trace}\left(\hat{\boldsymbol{A}}(n+1)\right)} \tag{54}$$

$$\boldsymbol{A}(n+1) = k_t\hat{\boldsymbol{A}}(n+1) \tag{55}$$

$$\boldsymbol{b}(n+1) = k_t\hat{\boldsymbol{b}}(n+1) \tag{56}$$

Time Update Equations

Table 2 Proposed Algorithm subroutine to calculate (17)

$$\delta_a = n(-2a_{1,2}(zp_t - 1) + z(a_{1,1} - 2a_{1,2}p_t) + 2a_{2,2}p_t(zp_t - 1) + za_{1,1}) \tag{63}$$

$$\delta_b = 4\left(a_{2,2}(zp_t - 1)^2 + z(za_{1,1} - 2a_{1,2}(zp_t - 1))\right)$$

$$\delta_1(z) = \frac{(n-4)(c_1 - c_2p_t)}{2(-c_2zp_t + c_1z + c_2)} - \frac{\delta_a}{\delta_b}$$

$$\delta_c = n(-2a_{1,2}(zp_t - 1) + z(a_{1,1} - 2a_{1,2}p_t) + 2a_{2,2}p_t(zp_t - 1) + za_{1,1})^2 \tag{64}$$

$$\delta_d = 4\left(a_{2,2}(zp_t - 1)^2 + z(za_{1,1} - 2a_{1,2}(zp_t - 1))\right)^2$$

$$\delta_2(z) = -\frac{n\left(2a_{2,2}p_t^2 - 4a_{1,2}p_t + 2a_{1,1}\right)}{4\left(a_{2,2}(zp_t - 1)^2 + z(za_{1,1} - 2a_{1,2}(zp_t - 1))\right)} + \frac{\delta_c}{\delta_d} - \frac{(n-4)(c_1 - c_2p_t)^2}{2(-c_2zp_t + c_1z + c_2)^2}$$

$$z_1 = z - \delta_1(z)/\delta_2(z) \tag{65}$$

$$p_z = -\delta_2(z) \tag{66}$$

$$a = \frac{\sqrt{p_z}(1 - z_1p_t)}{\sqrt{2}p_t}, \quad b = \frac{z_1\sqrt{p_z}}{\sqrt{2}}, \tag{67}$$

$q_w(n)$

$$\text{if } a + b < \delta \tag{68}$$

$$0$$

$$\text{elseif } |a| > 10 \wedge |b| > 10 \wedge \text{sign}(a)\langle 0 \wedge \text{sign}(b)\rangle 0$$

$$z_1 + \frac{2 - 2e^{-b^2 + a^2}}{\sqrt{2\pi p_z}\left(\frac{e^{-1/(2a^2)}}{\sqrt{\pi}a} + \frac{e^{-b^2 + a^2 - 1/(2b^2)}}{\sqrt{\pi}b}\right)}$$

$$\text{else if } |a| > 10 \wedge |b| > 10 \wedge \text{sign}(a) > 0 \wedge \text{sign}(b) < 0 \tag{69}$$

$$z_1 - \frac{2 - 2e^{-a^2 + b^2}}{\sqrt{2\pi p_z}\left(\frac{e^{-1/(2b^2)}}{\sqrt{\pi}b} + \frac{e^{-a^2 + b^2 - 1/(2a^2)}}{\sqrt{\pi}a}\right)}$$

$$\text{elseif } |a| > 3 \wedge |b| > 3 \wedge \text{sign}(a)\text{sign}(b) < 0 \tag{70}$$

$$z_1 + \frac{2e^{-a^2} - 2e^{-b^2}}{\sqrt{2\pi p_z}\{\text{sign}(a)\text{erfc}(|a|) + \text{sign}(b)\text{erfc}(|b|)\}}$$

else

$$z_1 + \frac{-2e^{-a^2} + 2e^{-b^2}}{\sqrt{2\pi p_z}\{\text{erf}(a) + \text{erf}(b)\}} \tag{71}$$

Simulation Results

This low value will make the **C**(n) matrix close to identity and the **A**(n) matrix close to singular, resulting in there being insufficient information to determine q_w and$q_v$accurately in a short time.

Simulation Results

Estimation of q_v and q_w . Average of 50 simulations

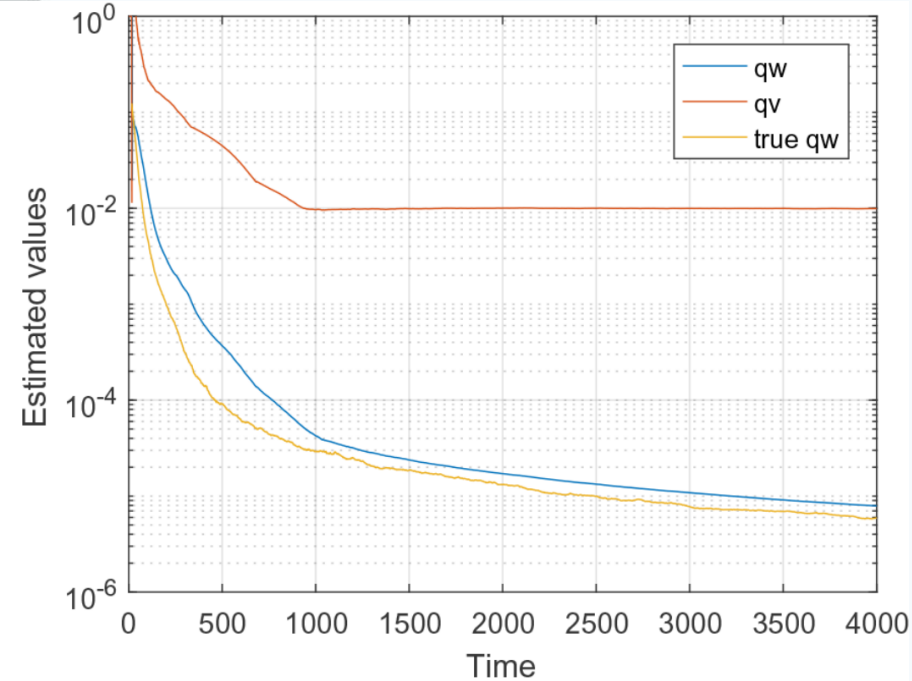


Table 3 General simulation parameters

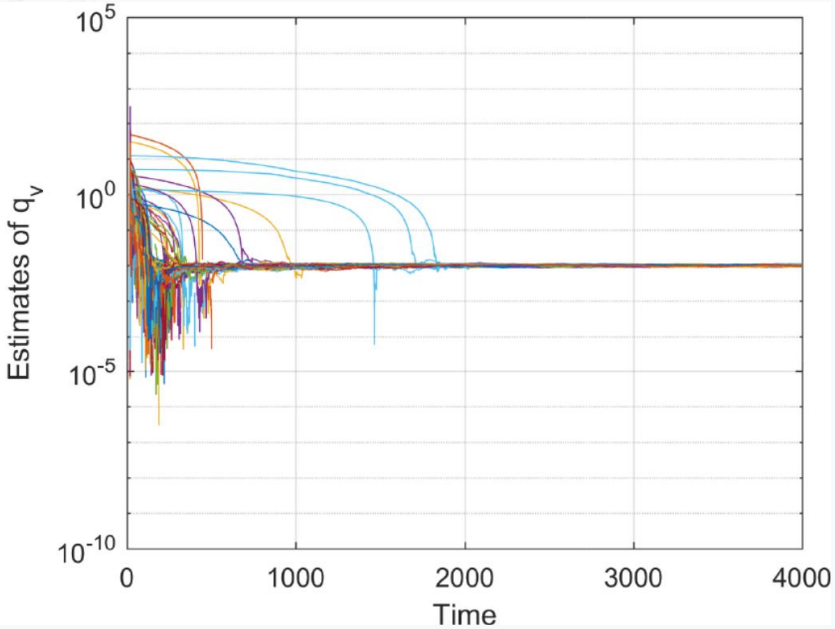
M	100	number of Monte Carlo runs
N	16	filter size
q_v	0.01	measurement noise variance
q_u	1.00	reference signal variance
s_w	1.00	coefficient standard deviation
$\boldsymbol{h}$	$[1, 0.5]$	reference filter impulse response

Table 4 General algorithm parameters

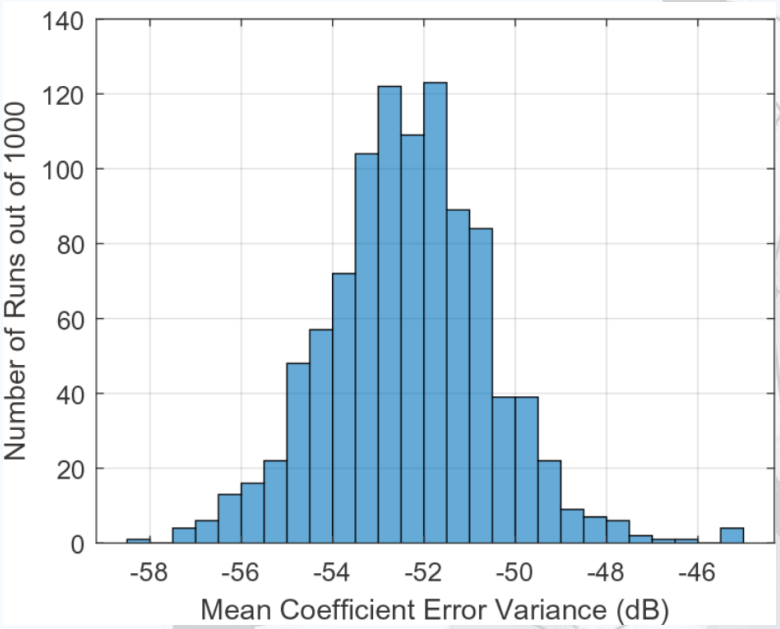
α	0.33	eigenvalue spread compensation factor
T	10^{11}	maximum trace of $\boldsymbol{A}$

Simulation Results

Estimates of q_v.Results of 100 single Monte Carlo runs.

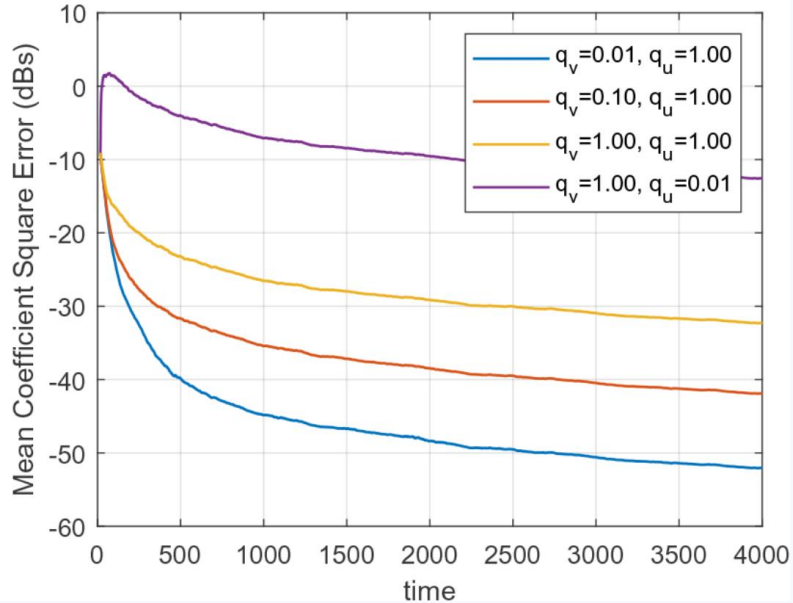


Histogram of the final value for the mean coefficient error variance

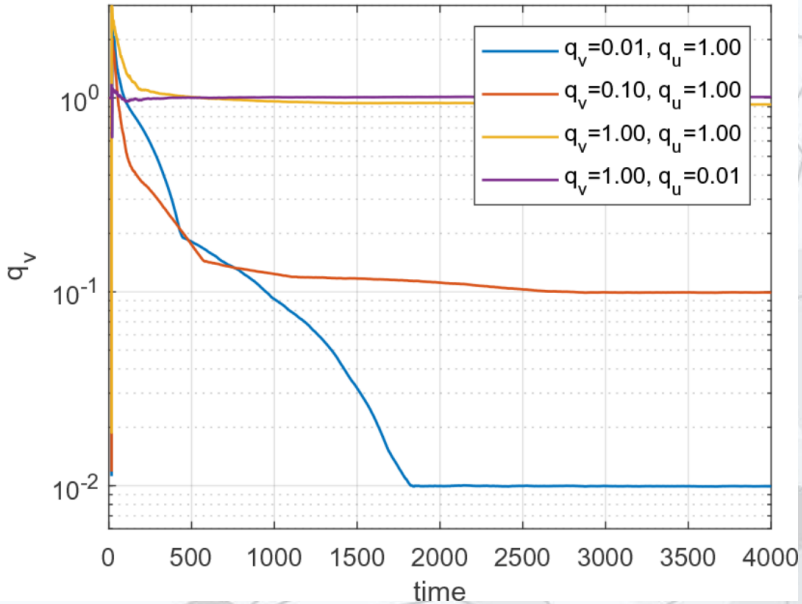


Simulation Results

Mean coefficient estimation error for different values of true- q_v and q_u.

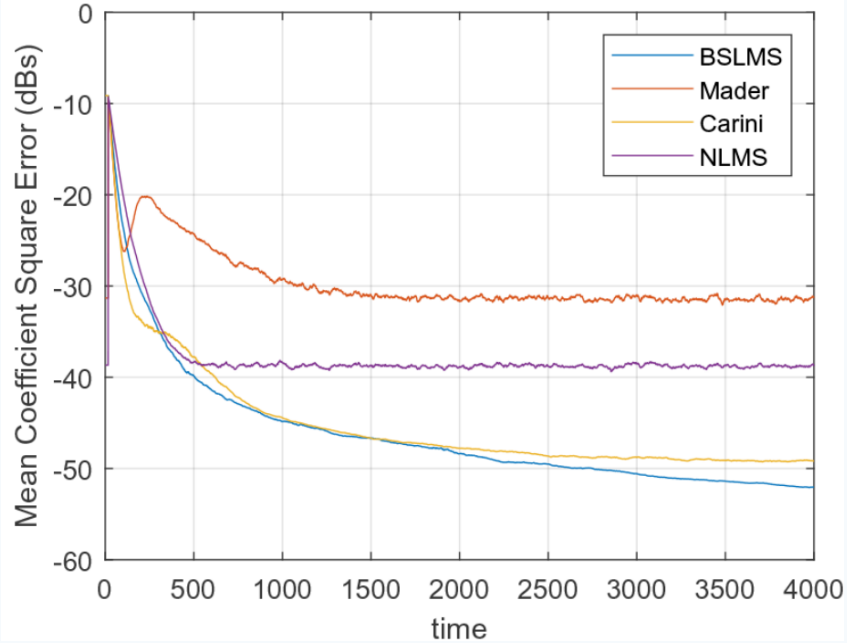


Estimate of the measurement noise variance q_v for different values of true- q_v and q_u.

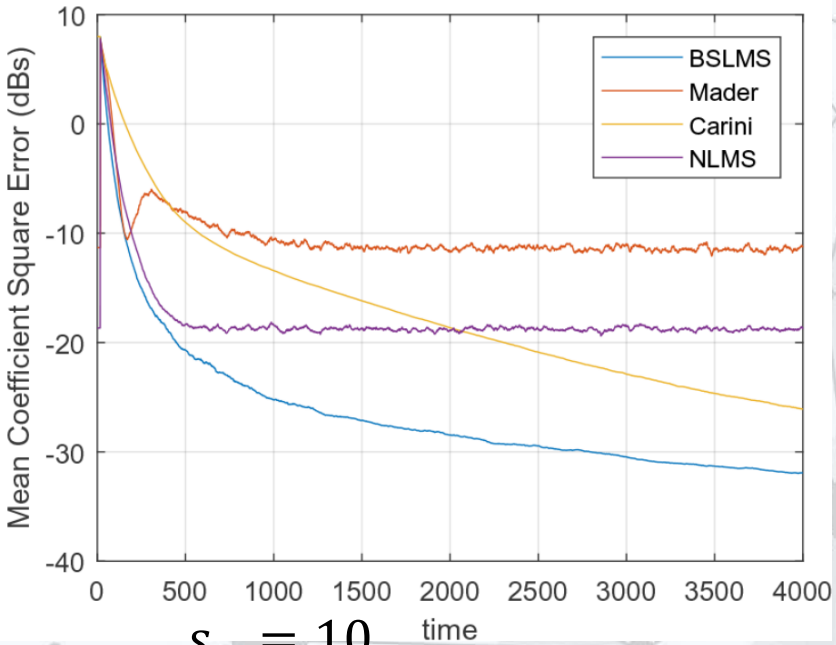


Simulation Results

Comparison with other algorithms with optimized parameters



Comparison with other algorithms with non-optimized parameters

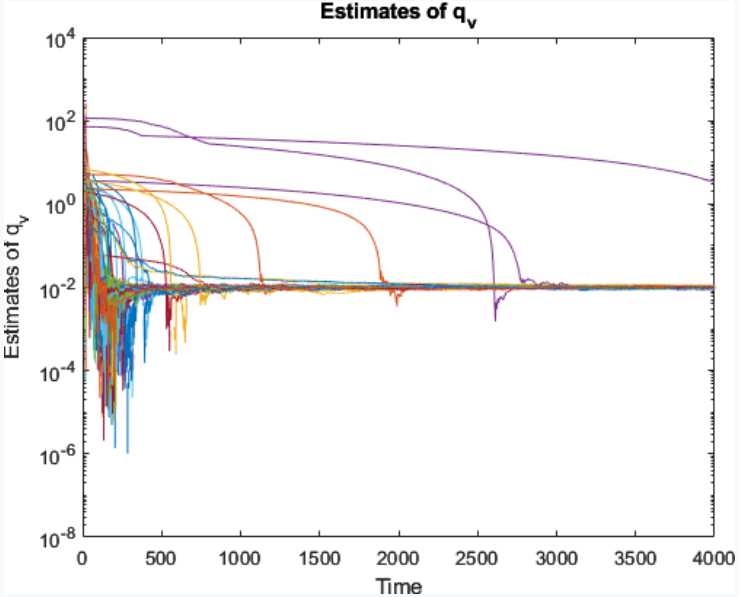


$s_w = 10$

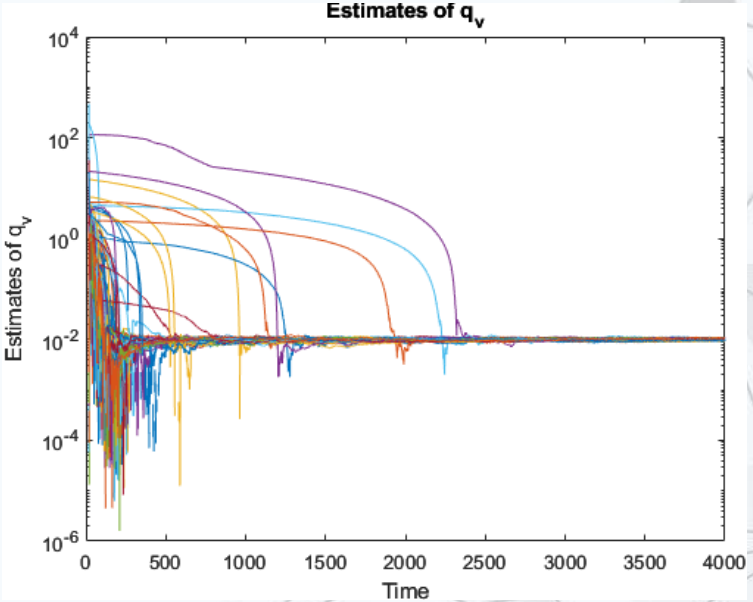
$q_v = 1$

Change Initial Value

Simulation results in paper

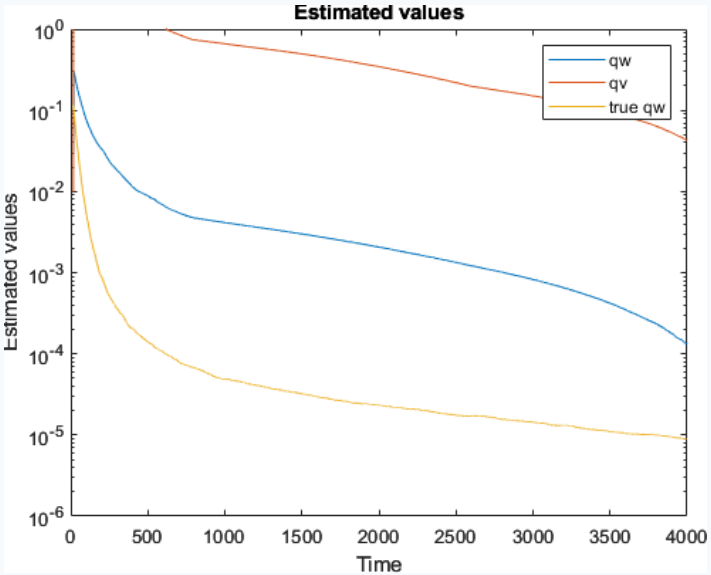


Our simulation results



Change Initial Value

Simulation results in paper



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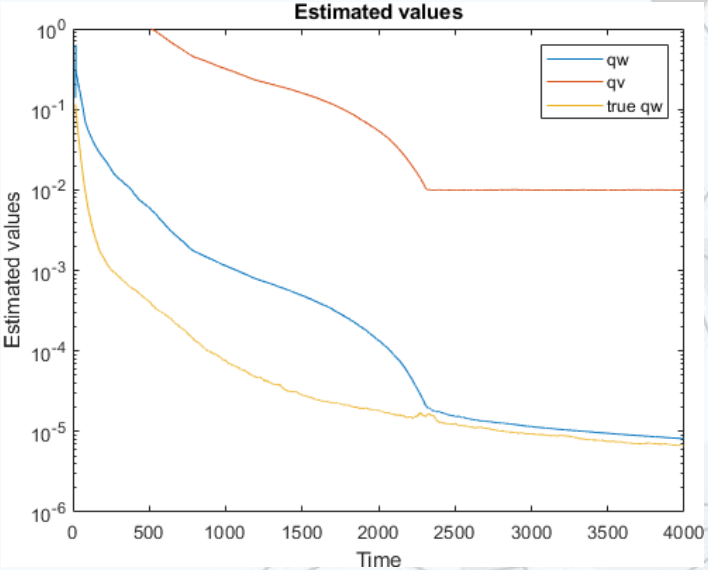

$$s_w = 10$$
$$q_v = 1$$

Conclusion

The step size

is selected using the expected value of the step of the KLMS algorithm by integrating the corresponding PDF.

Conclusion

- This presents an algorithm with excellent performance in system identification, namely, with a very small estimation error, thanks to the carefully selected step size of the LMS algorithm.
- To obtain estimates of coefficients error variance q_w and measurement noise variance q_v in a short time, it is required to model the decreases in q_v with the convergence of the filter.

End of My Presentation

Thank you for listening